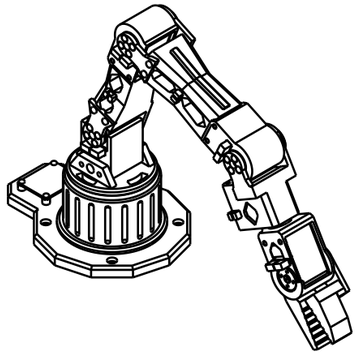
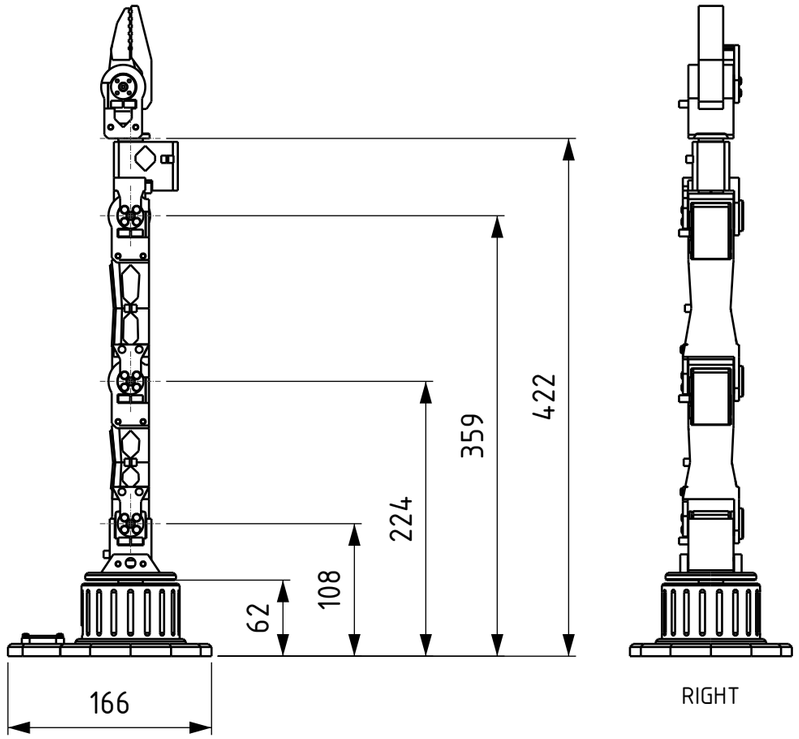


TOP



ISOMETRIC (NOT TO SCALE)



FRONT

RIGHT

PRINTED PARTS

PART NO.	NAME	QTY	SET
TAL-101	Base	1	A
TAL-102	Shoulder (turntable + cradle)	1	A
TAL-103	Upper arm	1	A
TAL-104	Forearm	1	B
TAL-105	Wrist	1	B
TAL-106	Gripper body	1	B
TAL-107	Moving jaw	1	B
TAL-108	Hub cap	3	A
TAL-110	Idler plate	2	B
TAL-111	Servo fit gauge (print first)	1	A

BOUGHT PARTS

ITEM	QTY
Feetech STS3215 servo (12 V)	6
Waveshare Bus Servo Adapter (A)	1
M3 x 6 button head (output and passive horns)	36
M3 x 10 button head (idler plates)	4
M2 x 5 self-tapping (servo cases)	12
M2 x 6 self-tapping (servo cases)	8
M2.5 x 6 self-tapping (board)	4
M4 bolts to the table (user's)	4
3-pin bus servo cables (with servos)	6

JOINT LAYOUT

1. Pan axis vertical through the base centre; pan horn face 62 above the table. Plinth 132 across corners.
2. Shoulder axis (along Y) 46 above the pan horn face. Upper arm 116, forearm 135 (axis to axis). Wrist flex axis to roll horn face 63.
3. Front view: zero pose (all joints 0). Isometric: 'ready' pose.
4. Assembly, each joint: plug a 3-pin cable into the servo; push the servo into its part along its entry path (part sheet notes); fit its M2 case screws; fit both horns (output horn centre screw; the passive horn sits loose on its boss); fit the next link with 4X M3 × 6 into each horn.
5. Print TAL-111 first and check the servo fit before printing the arm. Each part sheet gives the print orientation; print-ready 3MF files are in cad/print.

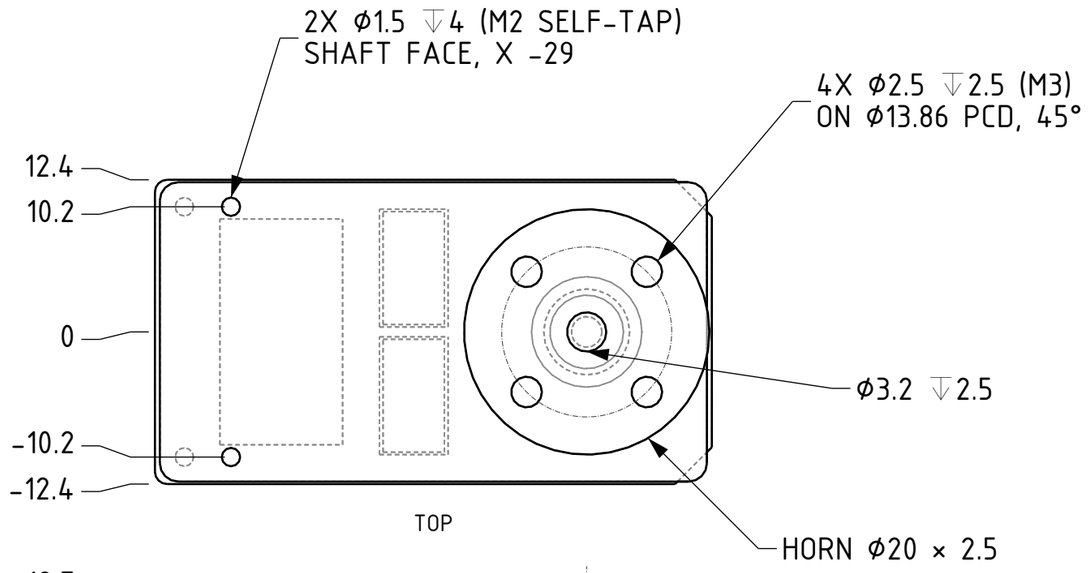
NOTES

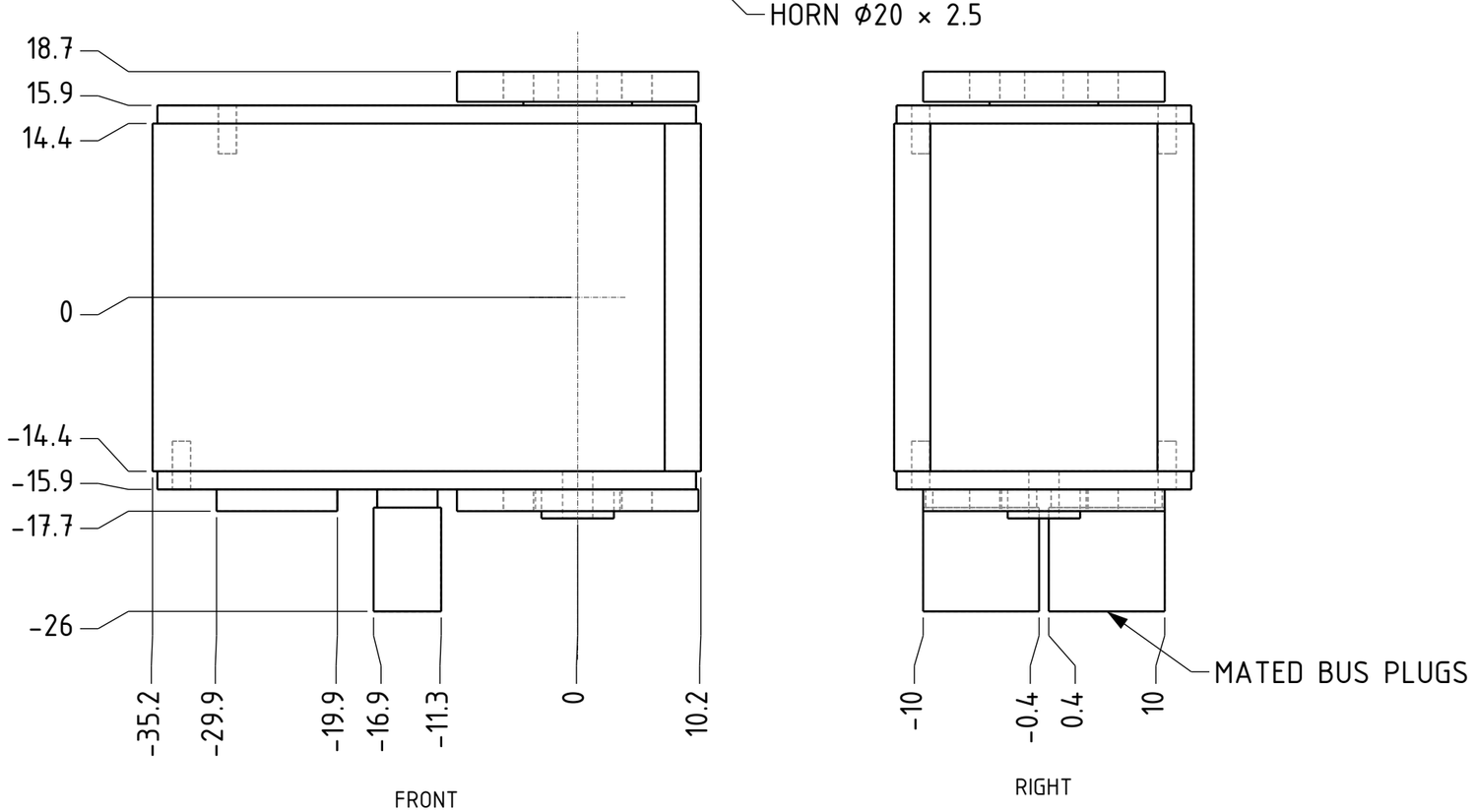
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part sheets give each part in its own frame; this sheet shows where those frames sit.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B

PROJECT TALOS 5-axis arm	EST. MASS -
PART General assembly and parts list	SCALE 1:5
PART NO. TAL-000	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL -	SHEET 1 of 21





SERVO (servo frame)

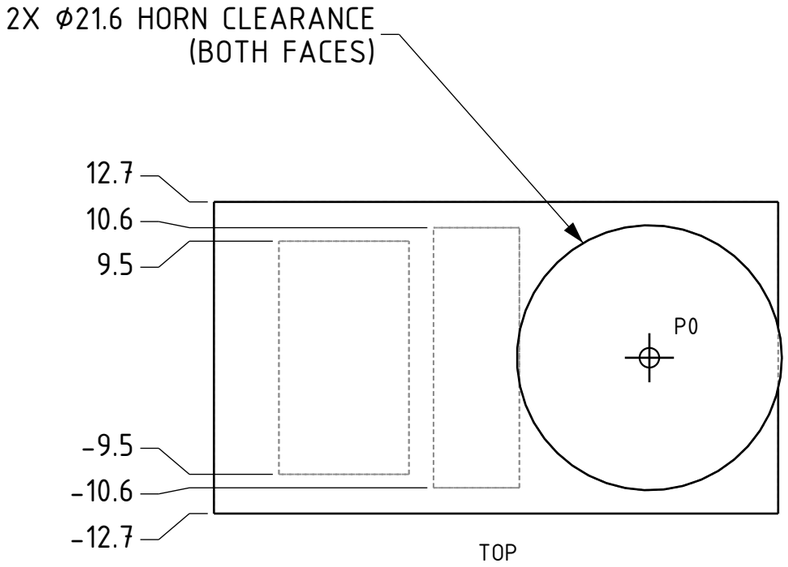
1. Body 45.4 × 24.8 × 28.8, x −35.2 to 10.2; +X end corners C3, −X end corners R1.
2. Caps 44.6 × 24.4 × 1.5, corners R1.5, centred on x −12.5, on both faces.
3. Back pad (connector face) x −29.9 to −19.9, y ±9.2, 1.8 high.
4. Bus headers x −16.6 to −11.6, y 0.6 to 9.8 and −9.8 to −0.6; mated plugs to z −26.
5. Output horn ⌀20 × 2.5 (z 16.2 to 18.7): 4X ⌀2.5 M3 on ⌀13.86 PCD at 45°, centre ⌀3.2.
6. Passive horn ⌀20 × 1.8 on the connector face, centre ⌀7.
7. Case holes ⌀1.5 ↧4 (M2 self-tap): shaft face at (−29, ±10.2), connector face at (−32.8, ±10.2).

NOTES

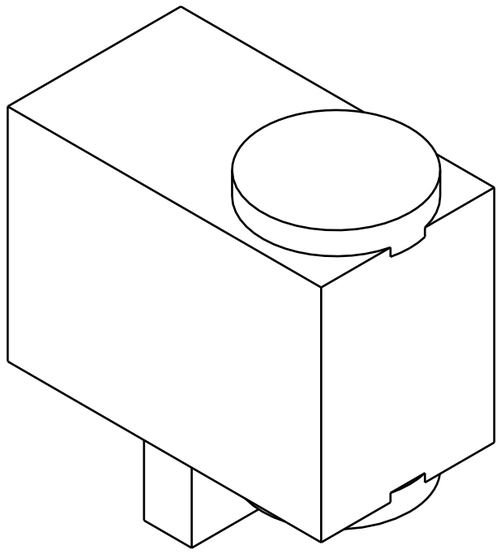
- Bought part: dimensions for reference, measured from the SO-ARM100 STS3215 model and Feetech's drawing. Check the connector position and case-hole depth on a real servo.
- Sheet 2 gives pocket P, the clearance envelope that every servo part cuts.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART STS3215 servo (reference) (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-06
MATERIAL bought part	SHEET 2 of 21



TOP



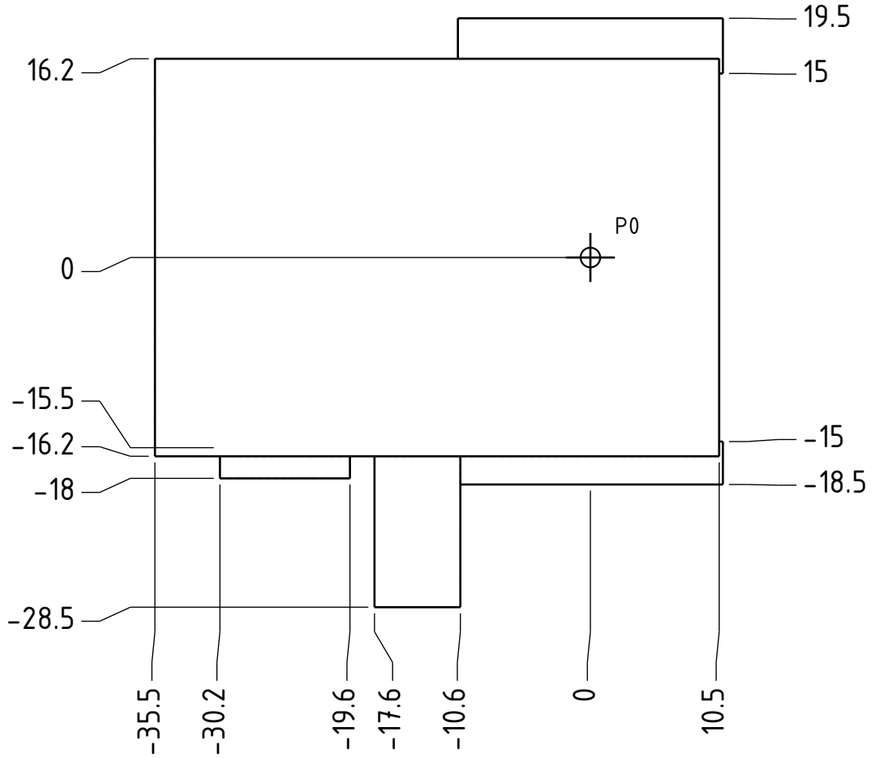
ISOMETRIC (NOT TO SCALE)

POCKET P (servo frame; 0.3 clearance included)

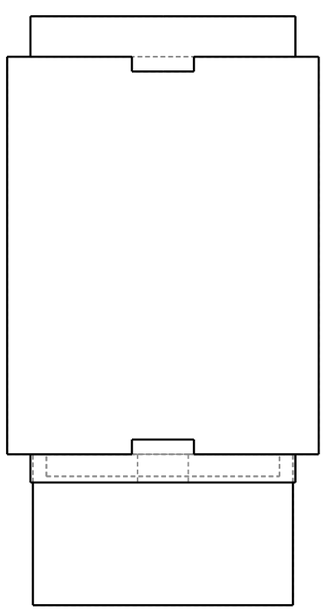
1. Main box x −35.5 to 10.5, y ±12.7, z ±16.2.
2. Back pad box x −30.2 to −19.6, y ±9.5, z −18.0 to −15.5.
3. Plug bay box x −17.6 to −10.6, y ±10.6, z −28.5 to −15.5 (mated plugs + cable bend).
4. Idler clearance cylinder ⌀21.6 on the shaft axis, z −18.5 to −15.0.
5. Horn clearance cylinder ⌀21.6 on the shaft axis, z 15.0 to 19.5.

NOTES

- Each part sheet locates P0 and gives the servo axes in the part frame; cut the union of the five bodies.
- All dimensions mm. Third-angle projection. Do not scale the drawing.



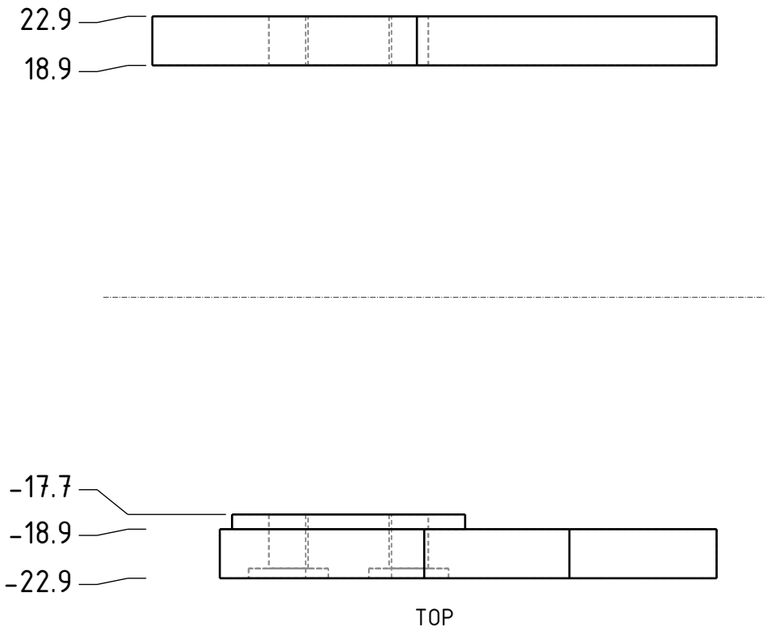
FRONT



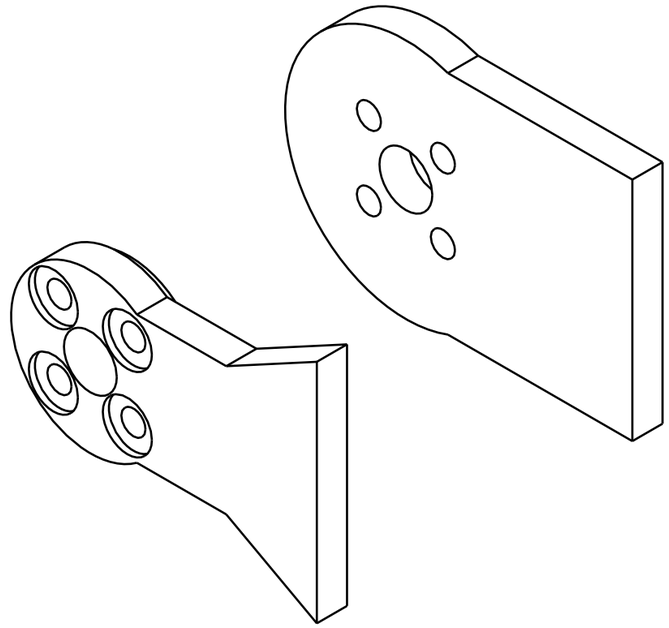
RIGHT

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Servo pocket P (cut into parts) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL -	SHEET 3 of 21



TOP



ISOMETRIC (NOT TO SCALE)

IDLER NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	0.6R = 18	±8.5
N4	R = 30	±H = ±15

CLEVIS (every pitch joint)

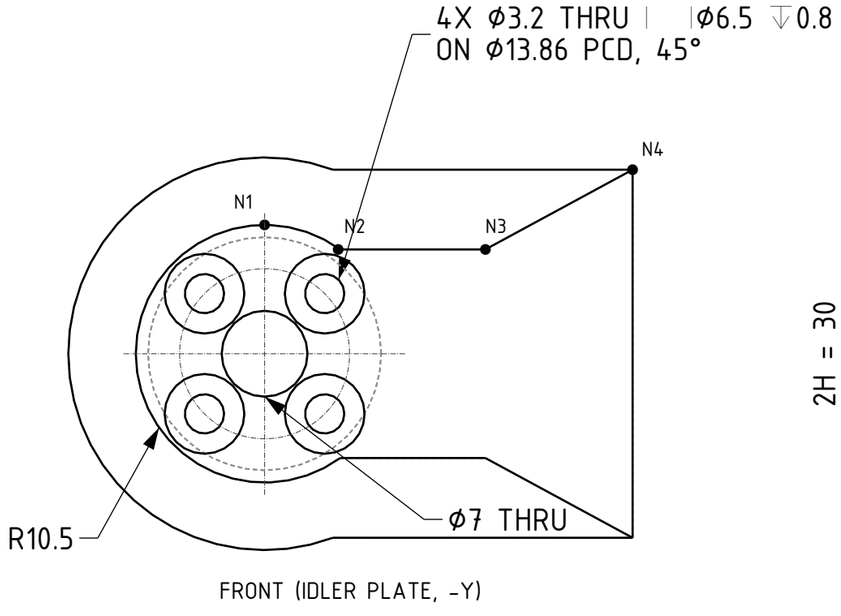
Drawn for the arm links: reach R = 30, half-height H = 15. The wrist uses R = 24, H = 12.

Not printed on its own: the plates are modelled into the links. On TAL-103 and TAL-104 the idler plate is the separate part TAL-110; on the wrist (TAL-105) both plates are wrist body.

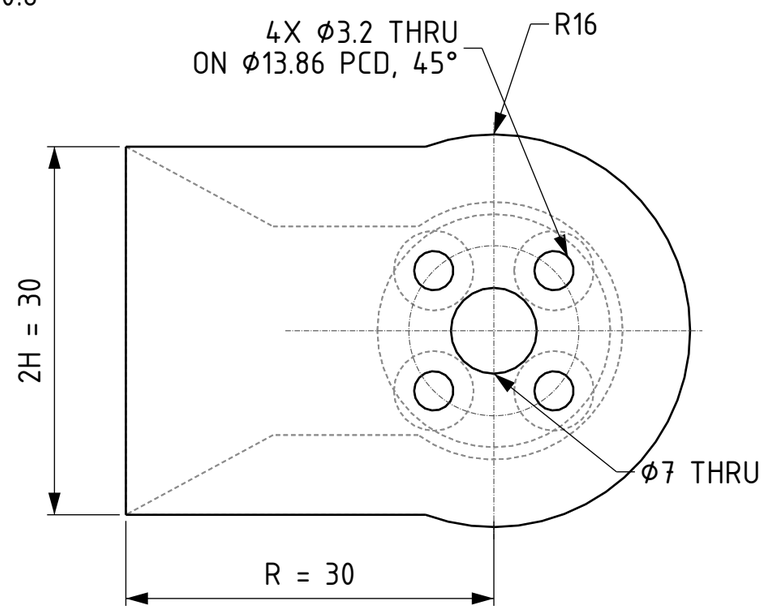
1. Output plate (+Y), y 18.9 to 22.9: circle R16 at the origin + rectangle x 0 to R, z ±H.
2. Output plate holes, through: Ø7 centre (horn spline screw) + 4X Ø3.2 on Ø13.86 PCD at 45°.
3. Idler plate (-Y), y -22.9 to -18.9: circle R10.5 + neck polygon N1 to N4 (mirrored). The neck clears the parent servo's bus plugs.
4. Idler plate pad Ø19 on the inner face, y -18.9 to -17.7, bearing on the passive horn; centre Ø7 through (clears the idler boss); 4X Ø3.2 on Ø13.86 PCD at 45°, counterbore Ø6.5 0.8 deep from outside (4X M3 × 6 into the passive horn, which turns on the boss).
5. Clevis gap in the link body: remove a cylinder R21 about Y between y ±18.9, and everything at x < 0 between those planes (dimensioned on each link's section).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.



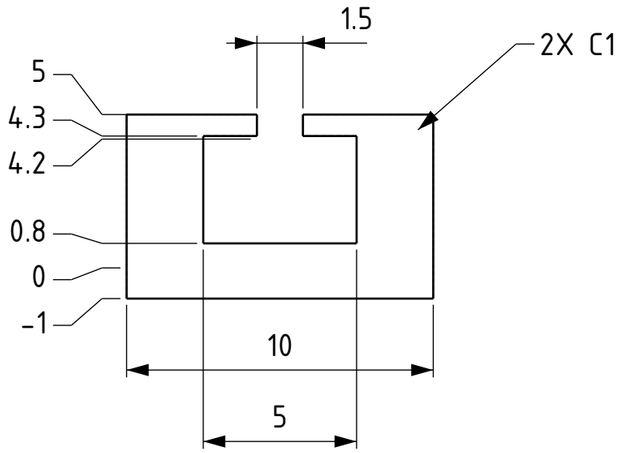
FRONT (IDLER PLATE, -Y)



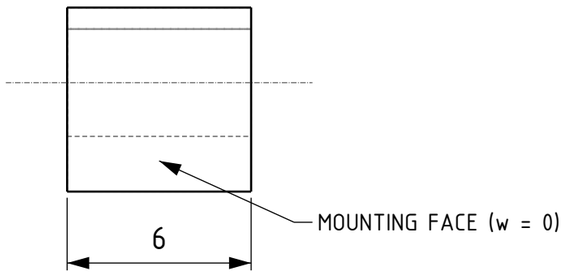
REAR (OUTPUT PLATE, +Y)

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clevis (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 21



CABLE CLIP: END VIEW



CABLE CLIP: SIDE VIEW

SEAT S (FROM THE SERVO MID-PLANE)

PART	S
arm shells, gripper	17.3
shoulder	18.5
base	18.4
wrist	18.0

COMMON FEATURES

CABLE CLIP, one bus cable. Local u along the cable, v across, w out of the mounting face.

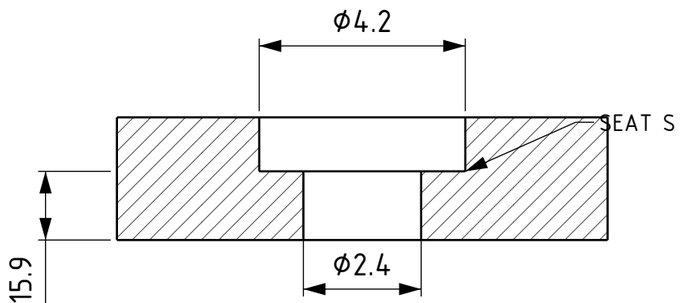
1. Block u ±3, v ±5, w -1 to 5 (sunk 1 into the face).
2. Tunnel through along u: v ±2.5, w 0.8 to 4.3. Slit v ±0.75 from w 4.2 out through the top.
3. Chamfer C1 on the two outer long edges.

CASE SCREW HOLE (M2 self-tap into the servo case): ⌀2.4 from the servo face to the seat S, then ⌀4.2 counterbore from S outward through the part.

HEX WINDOW H(xa, xb, zc, k): hexagon (xa, zc) (xa+k, zc-k) (xb-k, zc-k) (xb, zc) (xb-k, zc+k) (xa+k, zc+k), cut through the part along Y. Drawn for H(0, 40, 0, 10).

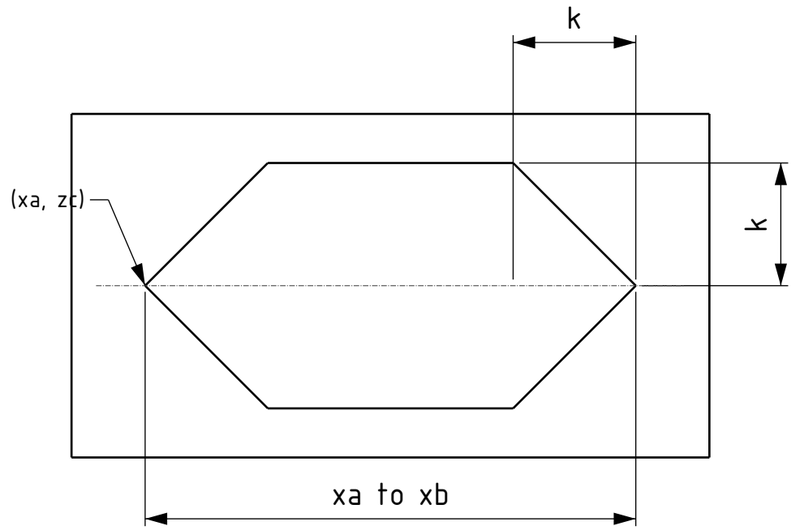
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



CASE SCREW HOLE (SECTION) SCALE 8:1

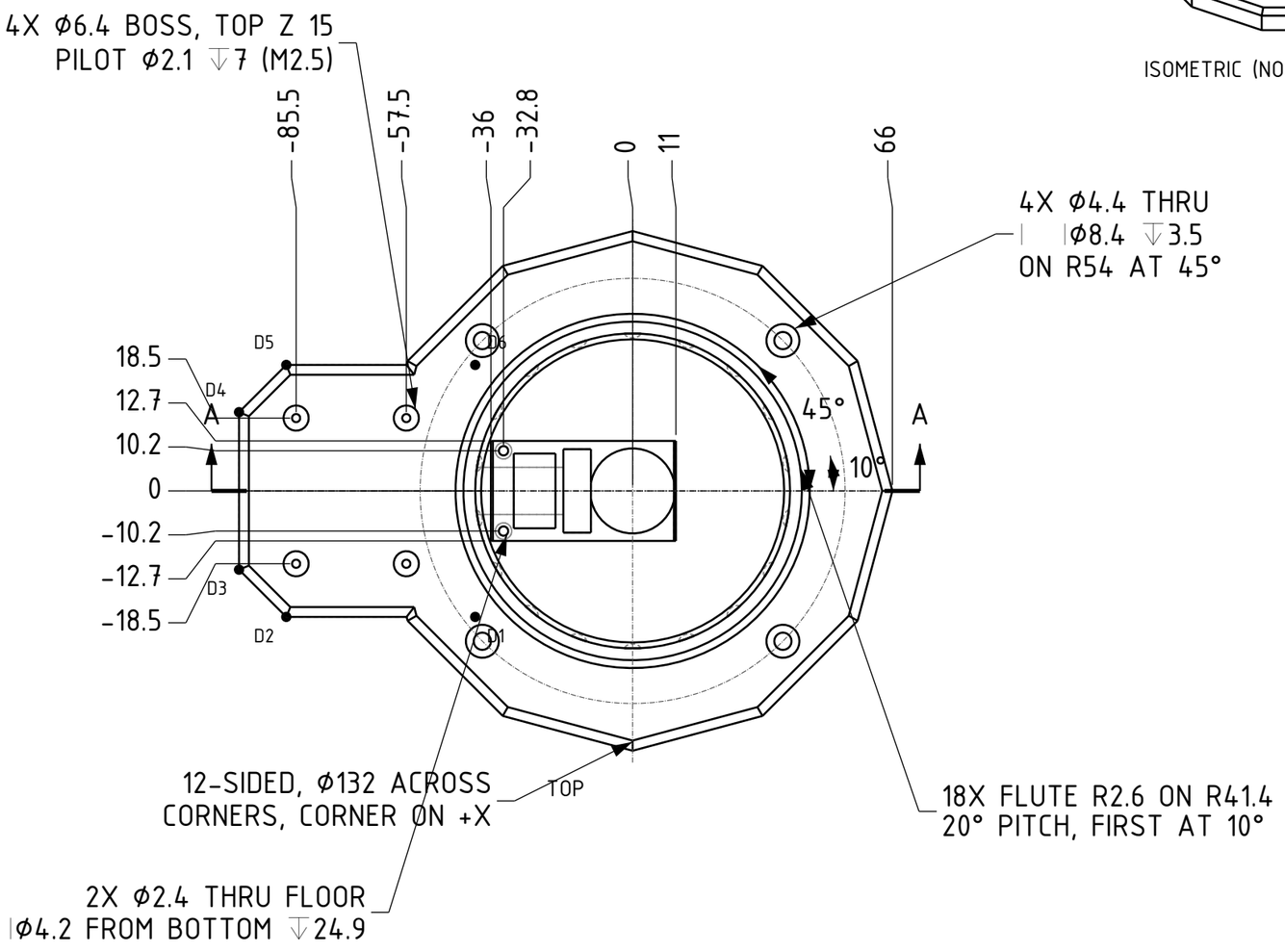
SERVO FACE (BOTTOM) IS 15.9
FROM THE SERVO MID-PLANE

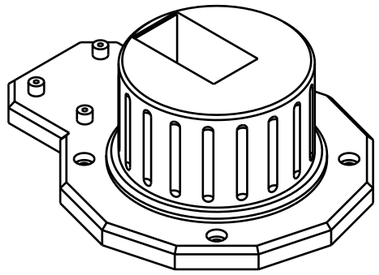


HEX WINDOW H(xa, xb, zc, k) SCALE 2:1

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clip, screw hole, window (part sheet 2 of 2)	SCALE 5:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 21





ISOMETRIC (NOT TO SCALE)

BOARD DECK POINTS (X, Y)

PT	X	Y
D1	-40	-32
D2	-88	-32
D3	-100	-20
D4	-100	20
D5	-88	32
D6	-40	32

FEATURES (in order)

1. Plinth: regular 12-gon, ⌀132 across corners (a corner on +X), z 0 to 10.
2. Board deck: polygon D1 to D6, z 0 to 10; union with the plinth.
3. Chamfer C2.5 on all top outer edges (z 10).
4. Mounting holes: 4X ⌀4.4 through on R54 at 45° + 90°n; counterbore ⌀8.4 from z 6.5 up.
5. Collar ⌀90, z 10 to 14, C2 on its top edge.
6. Column ⌀80, z 10 to 59.2, C1.5 on its top edge.
7. Board standoffs: 4X ⌀6.4, z 9 to 15, at x -57.5 / -85.5, y ±18.5; pilot ⌀2.1 from z 8 to 16.
8. Flutes: 18X vertical ⌀5.2 on R41.4 at 10° + 20°n, axis z 19 to 52, spherical ends; cut.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 43.3): servo +Z (shaft) along +Z, servo +X along +X, servo +Y along +Y.
10. Top opening: cut box x -36 to 11, y ±12.7, from z 58.2 up.
11. Cable tunnel (plug bay to board deck): cut box x -60 to -13, y ±6, z 14.8 to 24.5.
12. Case screw holes from below (TAL-002, seat 18.4) at x -32.8, y ±10.2: ⌀2.4 z 24.9 to 27.5, ⌀4.2 counterbore from the bottom to z 24.9.

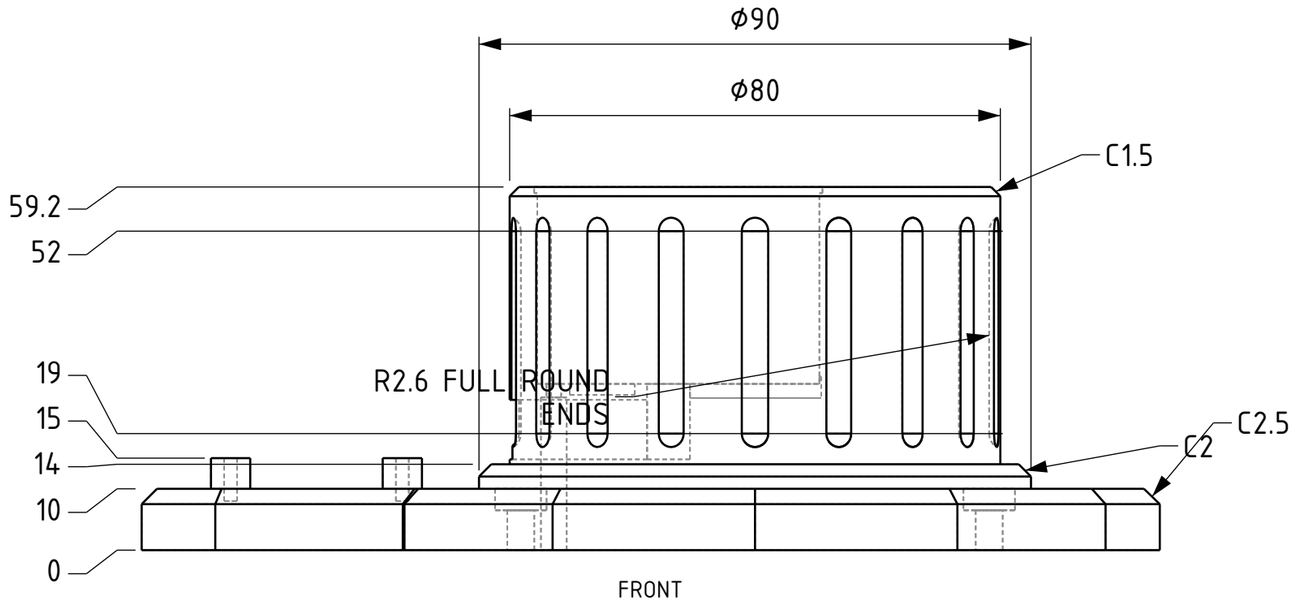
NOTES

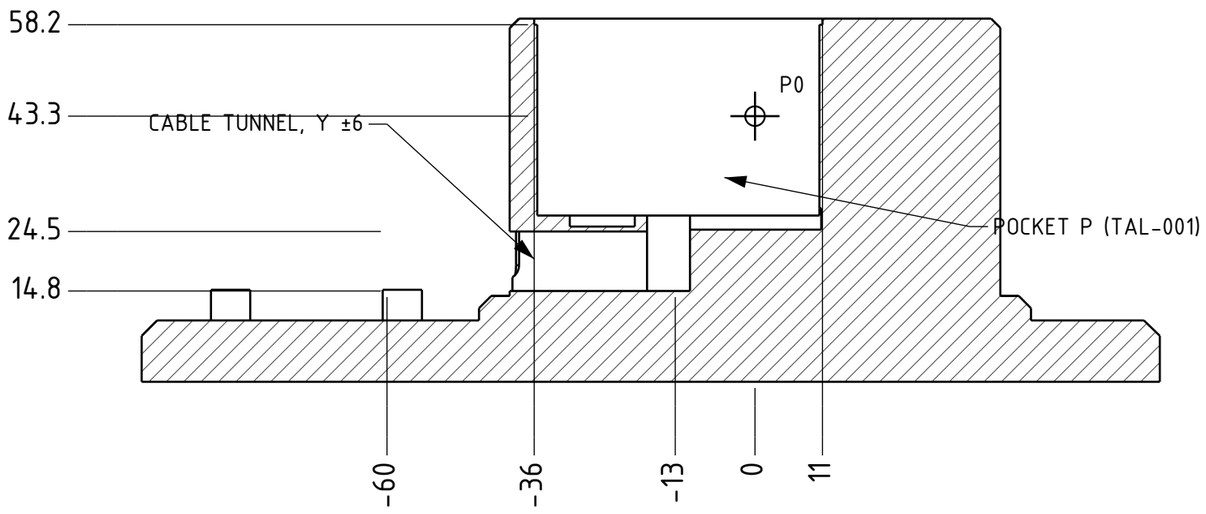
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top; its plugs hang into the bay under it.
- Print: plinth down; no support.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge

PROJECT TALOS 5-axis arm	EST. MASS 155 g
PART Base (part sheet 1 of 2)	SCALE 2:3
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 6 of 21





SECTION A-A

SECTION A-A

Section on the XZ plane (cutting line on sheet 1), looking toward +Y.

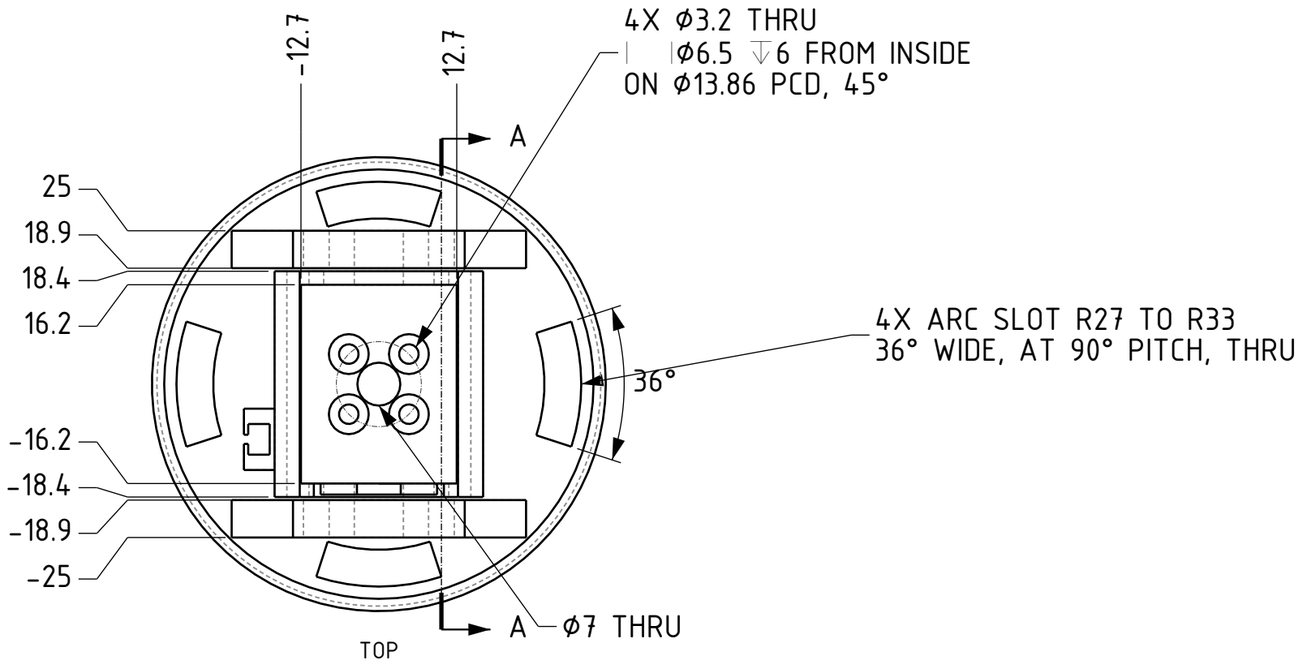
P0 = pocket P servo frame origin; servo axes along the part axes.

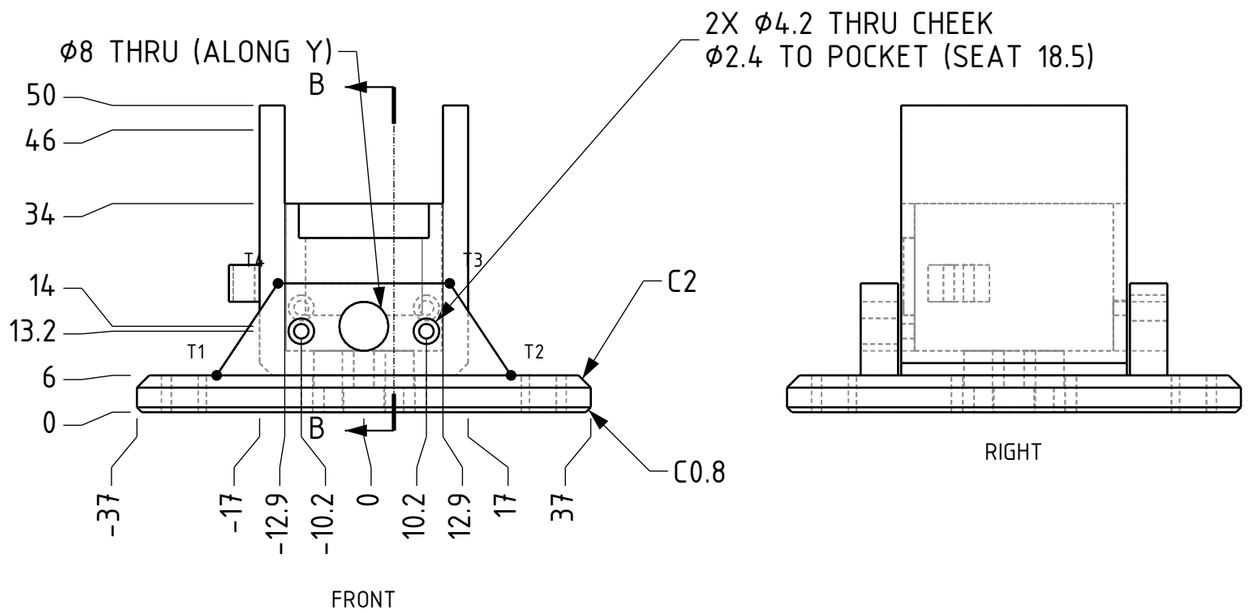
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Base (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 7 of 21





CHEEK PROFILE (X, Z)

PT	X	Z
T1	-24	6
T2	24	6
T3	14	21
T4	-14	21

FEATURES (in order)

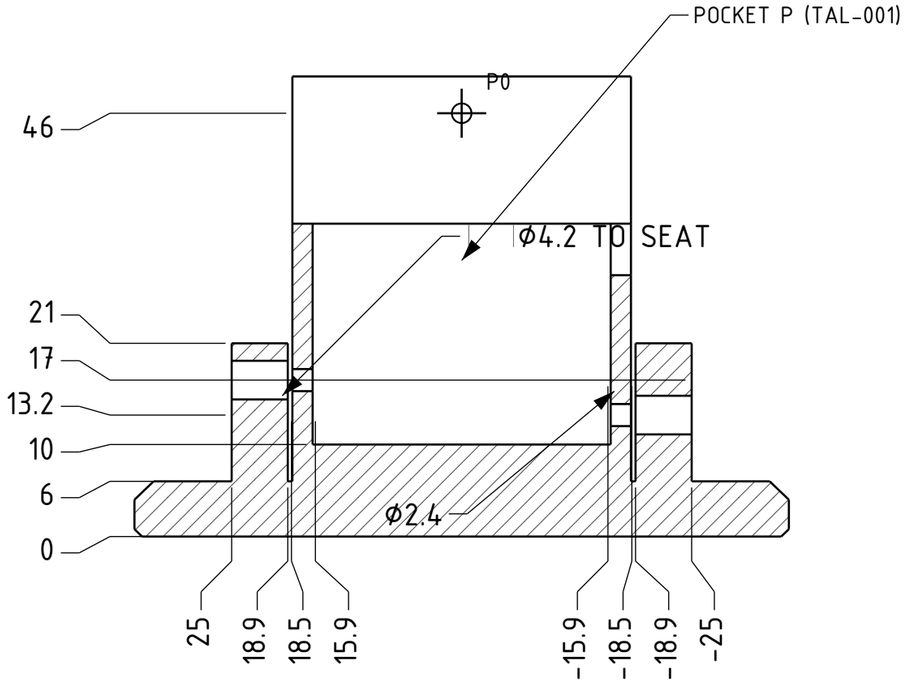
1. Turntable Ø74, z 0 to 6; C2 top edge, C0.8 bottom edge.
2. Cradle: box x ±17, y ±18.4, z 5 to 34.
3. Posts: boxes x -17 to -12.9 and 12.9 to 17, y ±18.4, z 30 to 50.
4. Cheeks: profile T1 to T4, y 18.9 to 25 and -25 to -18.9.
5. Cable clip (TAL-002) on the -X face: centre (-17, -9, 21), cable along Z.
6. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 46): servo +Z (shaft) along +Y, servo +X along +Z (body hangs down), servo +Y along +X.
7. Servo opening: cut box x ±12.7, y ±16.2, from z 10 up. Servo entry channels in the -Y cradle wall: back pad channel x ±9.5, y -18 to -16.2, from z 15.8 up; plug slot through the wall x ±10.6, from z 28.4 up.
8. Cable hole Ø8 along Y, through all, at x 0, z 14.
9. Pan horn: Ø7 centre through. 4X Ø3.2 through on Ø13.86 PCD at 45°, counterbore Ø6.5 from the floor (z 10) down to z 4 (M3 × 6 from inside the cradle).
10. Turntable windows: 4X arc slot R27 to R33, ±18° about 0°, 90°, 180°, 270°, through.
11. Case screw holes (TAL-002, seat 18.5) through the cheeks: +Y cheek at x ±10.2, z 17; -Y cheek at x ±10.2, z 13.2.

NOTES

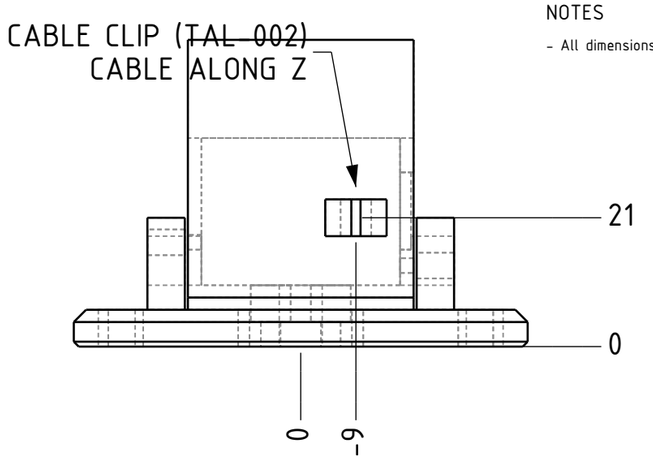
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the shoulder servo from the top, bottom end first with the shaft to +Y, before its horns and plugs; then 4X M2 × 6 case screws through the cheeks.
- Print: turntable down; no support.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 39 g
PART Shoulder (turntable + cradle) (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 8 of 21



SECTION A-A



LEFT SCALE 1:1

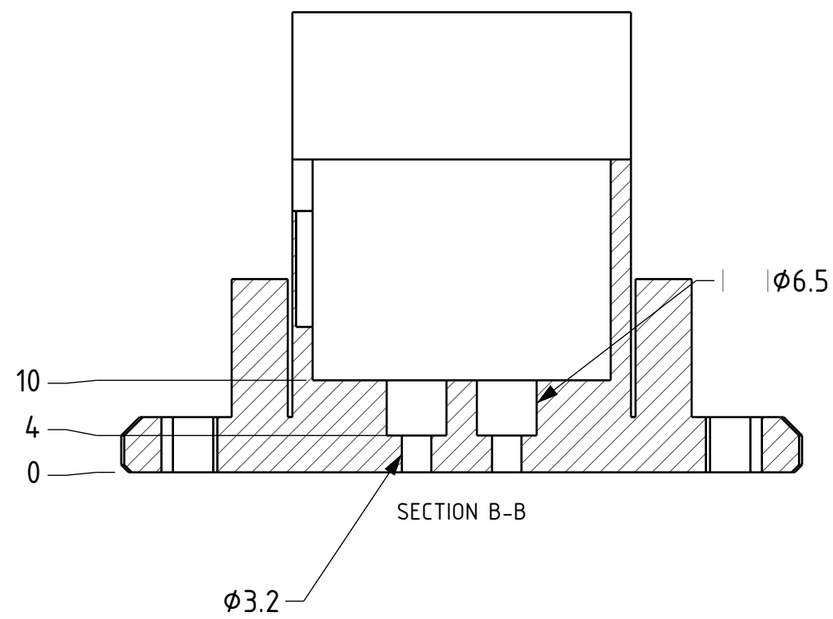
SECTIONS

A-A: plane x 10.2 through one hole of each case screw row (cutting line on sheet 1, top view), looking toward +X. The other holes mirror at x -10.2.

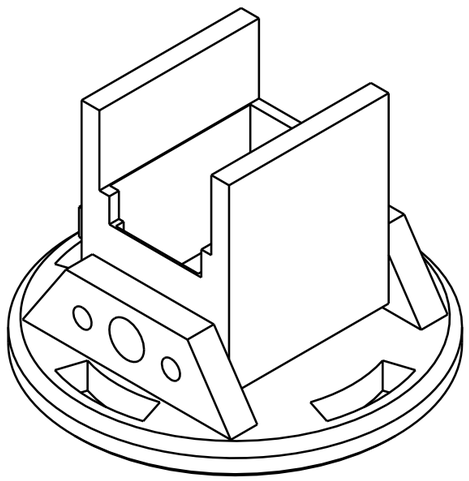
B-B: plane x 4.9 through two pan horn screw holes (cutting line on sheet 1, front view).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



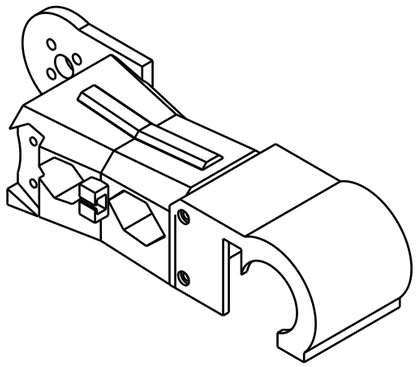
SECTION B-B



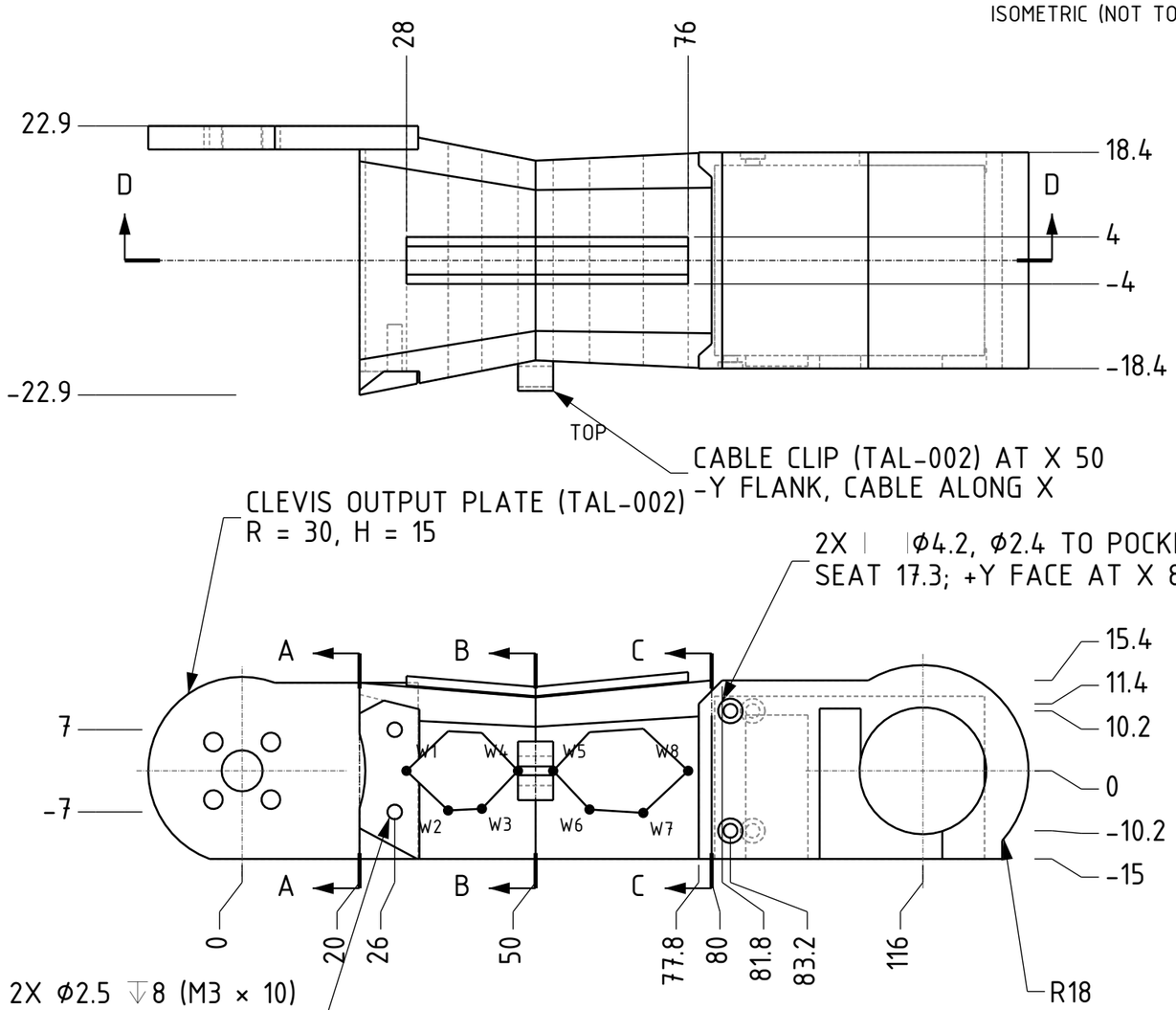
ISOMETRIC (NOT TO SCALE)

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Shoulder (turntable + cradle) (part sheet 2 of 2)	SCALE 3:2
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 9 of 21



ISOMETRIC (NOT TO SCALE)



WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	35.1	±6.75
W3	40.85	±6.46
W4	47	0
W5	53	0
W6	59.19	±6.58
W7	68.35	±7.16
W8	76	0

FEATURES (in order)

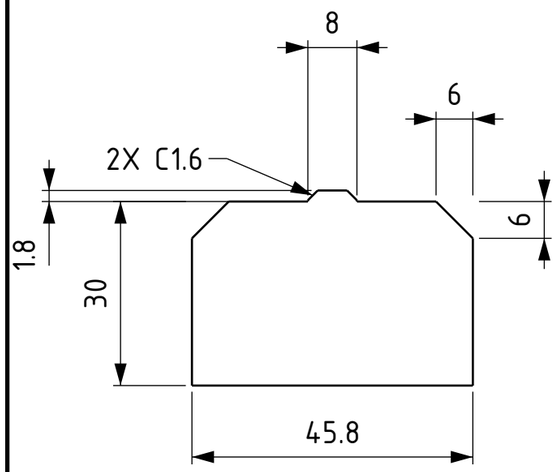
1. Clevis output plate only (TAL-002, +Y plate), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 50, 80. Every profile sits on z −15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (77.8, −15) (116, −15) (116, 15.4) (81.8, 15.4) (77.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 116.
4. Flat underside: cut everything below z −15 (trims the clevis hub and the R18 cylinder).
5. Nose floor: box x 125 to 129.5, y ±18.4, z −15 to −11; ties the shell walls together on the bed (the servo entry cut later trims its back to x 126.5).
6. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 76.
7. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
8. Cable clip (TAL-002) on the −Y flank at x 50, cable along X.
9. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes.
10. Pocket P (sheet TAL-001) with its servo frame origin P0 at (116, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along −Z.
11. Servo entry from below: cut box x 80.5 to 126.5, y ±16.2, z below 12.7 (opens the floor under the whole servo). In the −Y wall: back pad channel x 85.8 to 96.4, y −18 to −16.2, z below −9.5; notch through the wall x 98.4 to 119.3, z below 0 (joins the plug window and the idler hole).
12. Case screw holes (TAL-002, seat 17.3): +Y face at x 87, z ±10.2; −Y face at x 83.2, z ±10.2.
13. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.2, cut from y −23.9 to −18.9.
14. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y −18.9 8 deep (M3 × 10).

NOTES

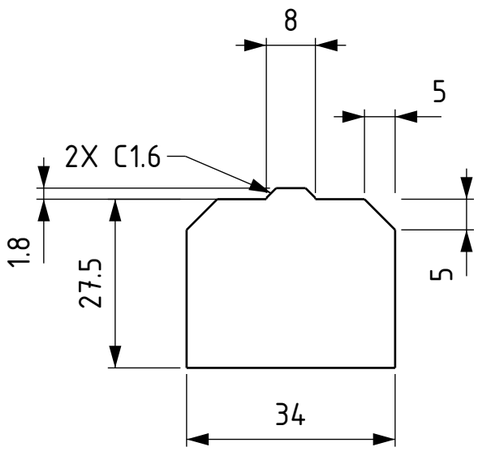
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/−0.2 unless noted; servo pocket clearances are already included.
- Fit the elbow servo from below, shaft to +Y, before its horns and plugs; then 4X M2 × 5 case screws.
- The elbow servo's bus plugs come out through the −Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z −15.

THIRD ANGLE PROJECTION

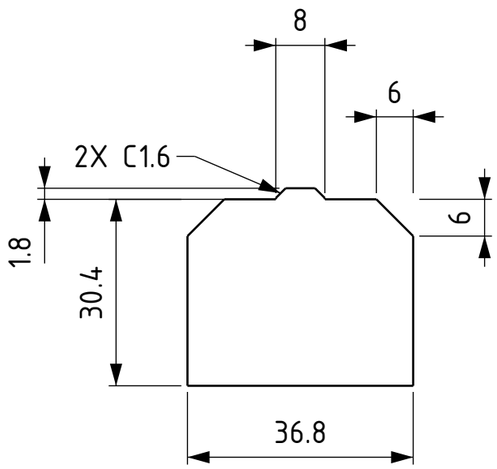
WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 48 g
PART Upper arm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 10 of 21



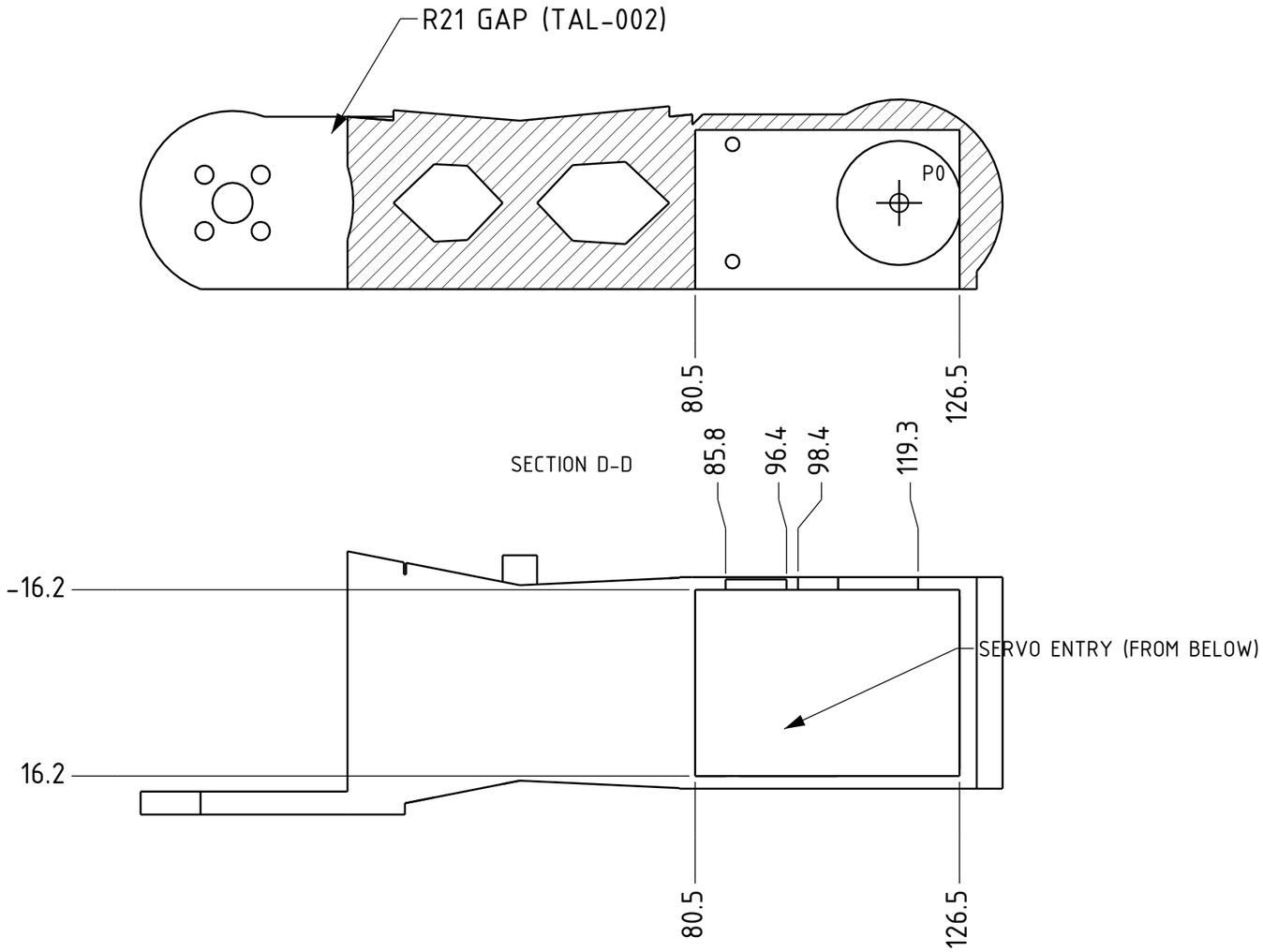
PROFILE A–A (X 20)



PROFILE B–B (X 50)



PROFILE C–C (X 80)



BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam section (top corners chamfered, underside flat and square-cornered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

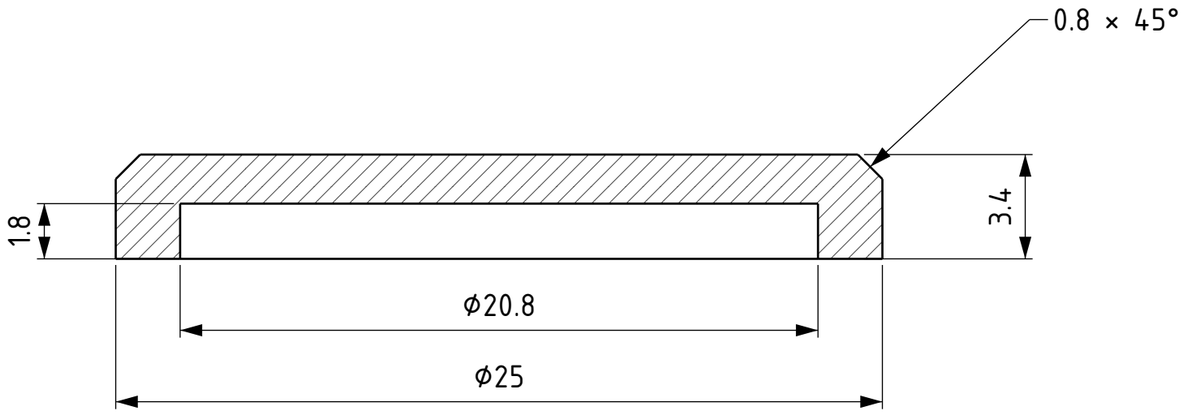
Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Upper arm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 11 of 21



SECTION A-A SCALE 5:1

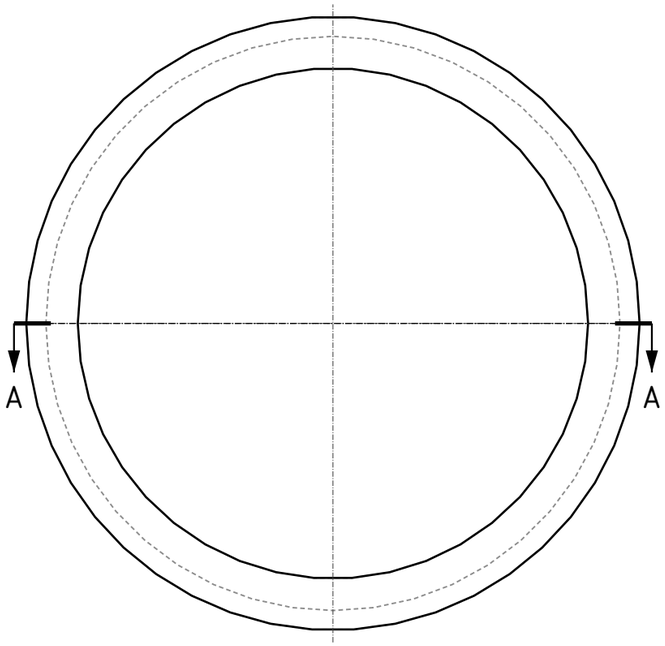
FEATURES (in order)

HUB CAP (output side, covers 4 horn screws):

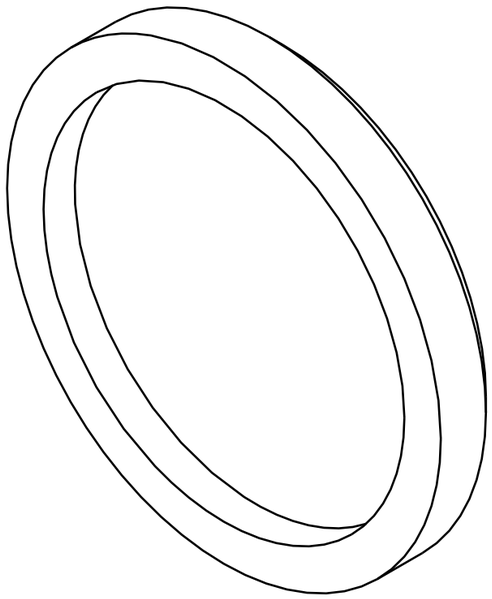
1. Disc ϕ25, 3.4 thick (y 0 to 3.4); chamfer 0.8 × 45° on the outer edge of the outer face.
2. Recess ϕ20.8, 1.8 deep from the inner face (clears M3 button heads, 1.65 high).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Press-fit or glue onto the output-side clevis plate over the four horn screws.
- Print: outer face down; no support.



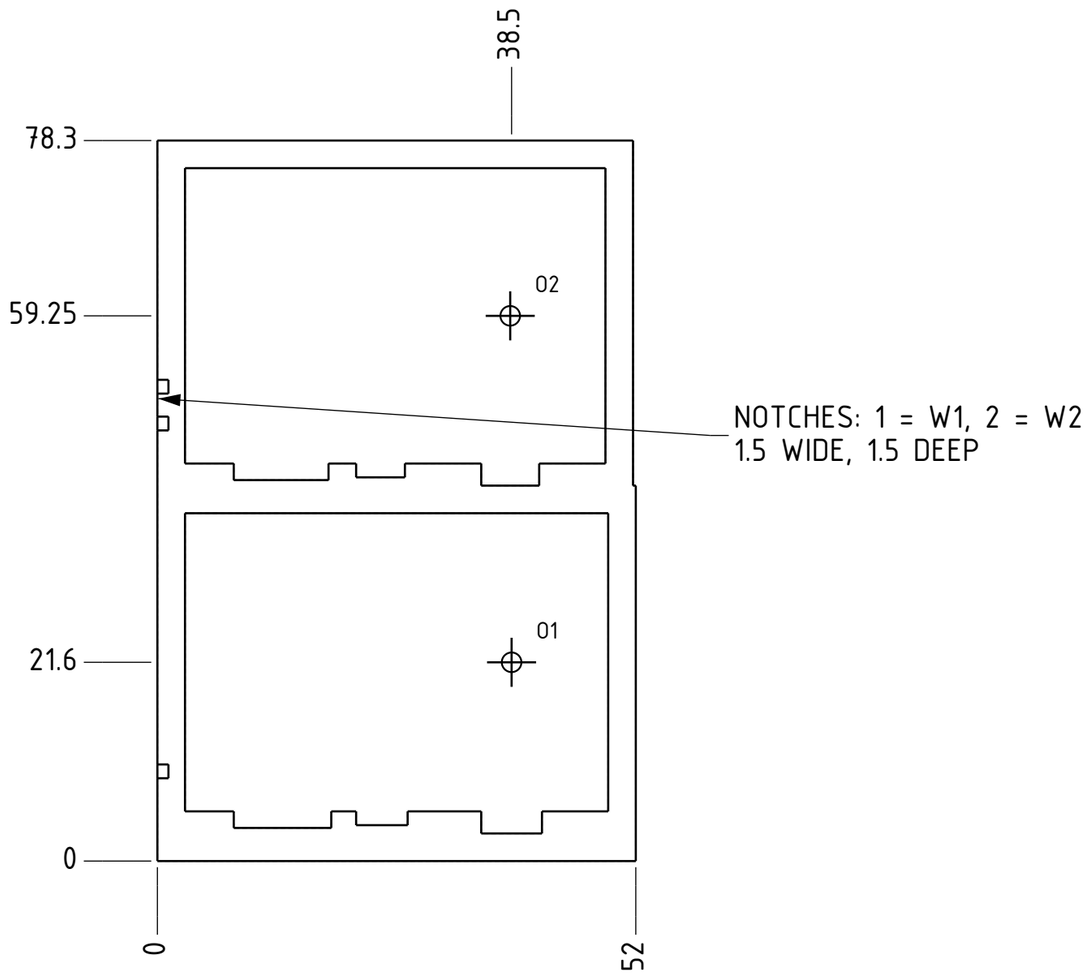
FRONT (INNER FACE)



ISOMETRIC (NOT TO SCALE)

THIRD ANGLE PROJECTION

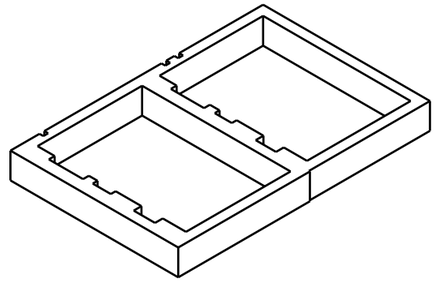
WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 1 g
PART Hub cap	SCALE 4:1
PART NO. TAL-108	UNITS / REV mm / A
QTY / ARM 3	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 19 of 21



TOP (AS PRINTED)



FRONT



ISOMETRIC (NOT TO SCALE)

GAUGE WINDOW AT ZERO CLEARANCE (X, Y)

RECTANGLE	X	Y
Body	−35.2 to 10.2	−15.9 to 15.9
Back pad notch	−29.9 to −19.9	−17.7 to 0
Header notch	−16.6 to −11.6	−17.4 to 0
Boss notch	−3 to 3	−18.3 to 0

FEATURES (in order)

1. Two frames, 8 thick (z 0 to 8), 3 walls, sharing the middle wall; overall 52 × 78.3.
2. Window W1 about O1 (38.5, 21.6): the four rectangles grown by 0.3, the clearance every servo pocket uses. Cut through Z.
3. Window W2 about O2 (38.35, 59.25): the same, grown by 0.15 (tight).
4. Notches on the x 0 wall's top edge: one beside W1, two beside W2.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Use: push the bare servo (no horns, no cables) through each window along its width, back face (connectors, idler boss) toward the notched side of the window.
- It should go through W1 by hand with light friction and not through W2. If W1 binds, or W2 lets it through, tune the printer (flow, horizontal expansion) before printing the arm.
- Print: flat; no support. Print it first, with the same settings as the arm.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 10 g
PART Servo fit gauge	SCALE 3:2
PART NO. TAL-111	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 21 of 21